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Studierichting: Informatica
Oefeningenreeks 4A nr. 42.6

$$\int \frac{x}{2x^2-3x+1}dx$$

$$2x^2-3x+1=0$$

$$\Delta = 9-8=1$$

$$x_1=\frac{3+1}{4}=1$$

$$x_2=\frac{3-1}{4}=\frac{1}{2}$$

$$=\int \frac{x}{(x-1)(2x-1)}dx$$

$$\frac{x}{(x-1)(2x-1)}=\frac{A}{(x-1)}+\frac{B}{(2x-1)}$$

$$\Rightarrow A=\frac{x}{(2x-1)}=1 \text{ met } x=1$$

$$\Rightarrow B=\frac{x}{(x-1)}=-1 \text{ met } x=\frac{1}{2}$$

$$=\int \frac{x}{(x-1)(2x-1)}dx=\int \frac{1}{(x-1)}dx-\int \frac{1}{(2x-1)}dx$$

$$=\ln|x-1|-\frac{1}{2}\ln|2x-1|+c$$

References